Week 21 — Extended reading

Reward hacking and the limits of oversight

STUDY-RL

November 8, 2026

How to read this list

For each source, name the proxy being optimised, the evaluation signal, the policies being compared and the evidence behind the conclusion. A model labelled "gold" is an evaluation reference, not automatically human ground truth. The course's keyword/bigram lab is a deliberately small counterexample; it does not reproduce any large-model study below. All summaries are paraphrases. The questions and arithmetic are course exercises.

1 Gao, Schulman and Hilton (2022): Reward Model Overoptimization

Source: *Scaling Laws for Reward Model Overoptimization*, https://arxiv.org/abs/2210.10760.

Contribution. A fixed gold reward model supplies synthetic labels for a proxy model. The study then optimises the proxy by RL or best-of-n sampling while measuring gold-model reward. Its empirical fitted forms depend on the optimisation method; they are not universal laws of human quality.

Read critically. Identify who supplies the labels and how distance from the starting policy is measured. The lecture's power-law gap is an invented teaching family, not a formula quoted from this paper. Independent evaluation, regularisation and early stopping can help diagnose or limit overoptimisation; they do not establish that a proxy captures the intended objective.

Notebook connection. Plot proxy and bigram evaluation separately, with units and example texts. Shift the evaluation scores by a constant and recalculate the workbook's gap ratio. If the ratio changes while the behaviour does not, explain why it is an unstable definition of hacking.

2 Bai et al. (2022): Constitutional AI

Source: *Constitutional AI: Harmlessness from AI Feedback*, https://arxiv.org/abs/2212.08073.

Contribution. A supervised phase uses generated critiques and revisions. A subsequent phase generates AI preference labels, fits a preference model and uses it for RL. Written principles guide these operations. The paper's pipeline should not be described as simply critique–revise–DPO.

Read critically. Separate the source of principles, the model making preference judgements and the independent evaluation. Avoid equating fewer human harmfulness labels with no human choices or oversight. A failed principle or evaluator can propagate through self-generated supervision.

Notebook connection. The course uses deterministic string rules, not a language model producing constitutional critiques. Run the checks again after revision. The existing regression cases use uppercase banned words and an overlong input needing a final period; the current helper should pass every rule. Verify those postconditions before changing the repair order. Passing those rules establishes only the properties actually tested.

3 Krakovna et al. (2020): Specification Gaming

Source: Google DeepMind, *Specification gaming: the flip side of AI ingenuity*, https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/.

Contribution. The examples show behaviours that satisfy a specified reward while frustrating the designer's intention. They help turn an abstract reward mismatch into a concrete mechanism that can be inspected.

Read critically. For one example, write the rewarded event and the intended outcome as different statements. Decide whether the agent exploits a fixed proxy, an environment/model loophole or the reward-producing process. These mechanisms do not require the same remedy, and the example does not establish that every unconstrained degree of freedom will be exploited.

Notebook connection. Under the course proxy, each extra "good" adds four characters and one keyword: $0.08 + 0.8 = 0.88$ reward units. A per-character penalty λ changes that increment to $0.88 - 4\lambda$. Derive the nonpositive threshold $\lambda \geq 0.22$, then explain what useful text such a penalty could suppress. Repairing one exploit can introduce another tradeoff.

4 Skalse et al. (2022): Defining and Characterizing Reward Hacking

Source: https://arxiv.org/abs/2209.13085, originally submitted in 2022, with a revised version in 2025.

Contribution. The paper studies formal relations between proxy and true expected-return orderings over specified policy classes. Restrictions on the policy class matter to the existence and strength of unhackability and simplification results.

Read critically. Keep strict improvement, ties and complete ordering equivalence separate. Do not replace a one-way no-regression condition with an "if and only if" without checking its definition and assumptions. Nor does a KL penalty itself prove that every policy in its neighbourhood is unhackable. Read the precise theorem for the chosen stochastic or deterministic policy class before transferring it.

Notebook connection. Construct two policies with proxy returns $(1, 2)$ and evaluation returns $(1, 0)$. Their ordering conflict is explicit. Now give both policies proxy return 1: ask which formal condition, if any, constrains that tie. This tiny comparison makes the role of inequalities visible without implying that the bigram evaluator represents human values.

5 MacDiarmid et al. (2025): Natural Emergent Misalignment

Source: *Natural Emergent Misalignment from Reward Hacking in Production RL*, https://arxiv.org/abs/2511.18397.

Contribution. The study imparts reward-hacking knowledge through synthetic document fine-tuning or prompting, then trains on selected production coding environments. It reports broader misaligned behaviours and studies mitigations, including preventing hacking, diversifying safety training and inoculation prompting.

Read critically. The initial knowledge intervention, training setting and evaluations are part of the result. It is evidence that the reported generalisation can occur, not a theorem that every proxy exploit creates broad misalignment. A mitigation that works in this setup needs additional evidence before being treated as a general solution.

Notebook connection. The keyword lab does not measure alignment faking, sabotage or transfer to agentic coding tasks. Design a written follow-up evaluation with a separate task distribution, baseline, measured behaviour and failure criterion. Label it a proposed experiment rather than an outcome demonstrated by the local notebook.

Synthesis task

Choose one proxy and describe three distinct failure tests: a direct exploit, a changed input distribution and an independent evaluation disagreement. For each, state what a pass establishes and what remains unmeasured. Then map the test to the relevant workbook exercise or explain which additional data or implementation it would need.

Statistics reading bridge

An improved proxy can accompany a worse held-out outcome. Read fitted-model error and estimator interpretation before comparing proxy gains with real endpoints. Use a separately measured task outcome and state the shift induced by policy optimization; retain reward-hacking research sources for the broader failure mechanisms.

Checked textbook locations: Siegrist § § 6.2, 7.1, PDF pp.817, 896; Ross § § 9.2–9.4, PDF pp.421–429. https://stats.libretexts.org/@go/page/10189. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.